

2. (1.75 points)

Given the following loss ratio triangle:

<u>Cumulative Reported Loss Ratios</u>					
Accident <u>Year</u>	12 <u>Months</u>	24 <u>Months</u>	36 <u>Months</u>	48 <u>Months</u>	60 <u>Months</u>
2010	3.0%	10.0%	15.7%	37.0%	37.0%
2011	5.1%	5.1%	25.0%	44.2%	48.0%
2012	2.5%	3.0%	40.0%	57.0%	59.2%
2013	1.6%	15.7%	22.2%	21.0%	
2014	0.0%	7.8%	16.7%		
2015	6.3%	12.4%			
2016	4.7%				

- Assume a tail factor of 1.15 from 60 months to ultimate.

Calculate the accident year 2014 ultimate loss ratio using the least squares method.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 2	
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE(S): A1A, A1B
SAMPLE ANSWERS	
<u>Sample 1</u> Need 2013 ultimate first: $X \text{ bar} = 1/3 * (0.37 + 0.442 + 0.57) = 0.4607$ $Y \text{ bar} = 1/3 * 1.15 * (0.37 + 0.48 + 0.592) = 0.5528$ $XY \text{ bar} = 1/3 * (0.37 * 1.15 * 0.37 + \dots) = 0.2632$ $X^2 \text{ bar} = 1/3 * (0.37^2 + \dots) = 0.2191$ $b = (XY \text{ bar} - X \text{ bar} * Y \text{ bar}) / (X^2 \text{ bar} - (X \text{ bar})^2) = 1.2435$ $a = Y \text{ bar} - b * X \text{ bar} = -0.0201$ Since $a < 0$, using link ratio method instead $2013 \text{ ultimate} = 0.21 * 1.15 * (0.37 + 0.48 + 0.592) / (0.37 + 0.442 + 0.57) = 0.2520$ Calculate 2014 ultimate $X \text{ bar} = 1/4 * (0.157 + 0.25 + 0.4 + 0.222) = 0.2573$ $Y \text{ bar} = 1/4 * (0.37 * 1.15 + 0.48 * 1.15 + 0.592 * 1.15 + 0.2520) = 0.4776$ $XY \text{ bar} = 1/4 * (0.157 * (0.37 * 1.15) + \dots) = 0.1333$ $X^2 \text{ bar} = 1/4 * (0.157^2 + \dots) = 0.0741$ $b = (XY \text{ bar} - X \text{ bar} * Y \text{ bar}) / (X^2 \text{ bar} - (X \text{ bar})^2) = 1.3187$ $a = Y \text{ bar} - b * X \text{ bar} = 0.1383$ $2014 \text{ ultimate} = a + b * 0.167 = 0.359$ <u>Sample 2</u> AY Ult L/R $10 0.4255 = 0.37 * 1.15$ $11 0.552 = 0.48 * 1.15$ $12 0.6808 = 0.592 * 1.15$ $b = (XY \text{ bar} - X \text{ bar} * Y \text{ bar}) / (X^2 \text{ bar} - (X \text{ bar})^2)$ $Y \text{ bar} = a + b * X \text{ bar}$ 2013 Ult L/R: $X \text{ bar} = (1.37 + 0.442 + 0.57) / 3 = 0.4607$ $Y \text{ bar} = (0.4255 + \dots + 0.6808) / 3 = 0.5528$ $XY \text{ bar} = (0.37 * 0.4255 + \dots + 0.57 * 0.6808) / 3 = 0.2632$ $X^2 \text{ bar} = (0.37^2 + \dots + 0.57^2) / 3 = 0.2191$ $b = 1.244 \quad a = -0.02$ Even though a negative "a" value could lead to impractical y values, I will use Least Squares since the problem says to, even though CL maybe more appropriate. $AY \text{ 2013 Ult L/R} = 1.244 * 0.21 - 0.02 = 0.2412$ AY 2014 Ult L/R: $X \text{ bar} = (0.157 + \dots + 0.222) / 4 = 0.2573$ $Y \text{ bar} = (0.4255 + \dots + 0.2412) / 4 = 0.4749$ $XY \text{ bar} = (0.157 * 0.4255 + \dots) / 4 = 0.1327$	

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$\bar{X^2} = (0.157^2 + \dots + 0.222^2)/4 = 0.0741$$

$$\text{AY 2014 Ult L/R} = 1.3307 \cdot 0.167 + 0.1325 = 0.335$$

EXAMINER'S REPORT

Candidates were expected to be aware that they cannot use a negative «a» parameter, because this could result in negative loss ratio.

Common errors include:

- Using a negative «a» parameter, without mentioning that the chain ladder method should be used instead.
- Not using as much data as possible in the calculation. This includes the 2013 data at 48 months.